A full fixed-K Barnes–Bessel limit with an integrated first correction

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Abstract

An exact unit-shift recurrence controls the positive horizontal tail of the recoil Barnes kernel uniformly in the mass. A complementary vertical-tail estimate permits a full fixed-K, α limit to the two-branch Bessel integral. A logarithmically growing truncation then justifies the first integrated correction with an absolute $O(\nu^{-2})$ remainder. Constants depend on K, α. This addresses the global gap in the preceding note, without asserting a joint growing-K limit or a physical finite-G four-point function.

1 Object and fixed-parameter statement

Set the de Sitter radius to one, $u = v = 0$, $K > 0$, real α, $\omega = i\alpha$, and $M = \nu^2/(2K)$. Our object is the internal chiral Barnes amplitude after the earlier fixed-mass $\kappa \to \infty$ limit. Physical state, algebra, external insertions and normalization remain as yet unidentified. Define $t = z + i(M + \nu)$, $v_0 = \nu/2$, and

$$b = \tfrac{1}{2} - i(M - \nu), \quad c = \tfrac{1}{2} - i(M + \nu), \quad C_{M,\nu} = e^{-\pi(M+\nu)} \frac{\Gamma(-iv_0)^4 \Gamma(b)}{\Gamma(c)}.$$

The normalized exact kernel is

$$Q_\nu(t) = e^{-i\pi t} \frac{\Gamma(t - iv_0)^4}{\Gamma(-iv_0)^4} \frac{\Gamma(b - t)}{\Gamma(b)} \frac{\Gamma(c)}{\Gamma(c + t)} \Gamma(\tfrac{1}{2} - t + \alpha/2)\Gamma(\tfrac{1}{2} - t - \alpha/2). \tag{1}$$

Fix $\eta > 0$, and an integer $N > |\alpha|/2 + 1/2$. Let $\mathcal{C}$ run from $-N - i\infty$ upward to $-N + i\eta$, then right to $+\infty + i\eta$. For sufficiently large ν, this is the pole-free deformation of the original Barnes contour. Its left ray points upward. Put

$$I_\nu = \frac{1}{2\pi i} \int_{\mathcal{C}} Q_\nu(t)\, \mathrm{d}t = \frac{\mathfrak{B}}{C_{M,\nu}}, \qquad J = -K\big[e^{i\pi\alpha} K_\alpha(K) + i\pi I_\alpha(K)\big].$$

Theorem 1. *For fixed $K > 0$, real α, as $\nu \to \infty$,*

$$I_\nu = J\left[1 + \frac{i}{\nu}(K^2 + \alpha^2 - 1)\right] + O_{K,\alpha}(\nu^{-2}). \tag{2}$$

The error is absolute, including when $J = 0$.

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2 The exact recurrence gives a uniform horizontal tail

Gamma recurrence [1], applied directly to (1), yields

$$\frac{Q_\nu(t+1)}{Q_\nu(t)} = R_\nu(t) := \frac{(t-i\nu/2)^4}{[(t+\frac{1}{2}-i\nu)^2+M^2][(t+\frac{1}{2})^2-\alpha^2/4]}. \tag{3}$$

The minus sign from $e^{-i\pi(t+1)}$ cancels the sign from $\Gamma(b-t-1)/\Gamma(b-t)$. This avoids using a fixed-mass Stirling constant as though it were mass-uniform.

Set

$$d = \max(\alpha^2/4-\eta^2, 0), \quad A = \max(2, 2K, 2d), \quad a = \nu/2-\eta, \quad L_\nu = M-\nu+\eta.$$

For $\nu \geq \max(8K, 4\eta, 2)$, $a \geq 0$, $L_\nu \geq M/2$. For $t = x+i\eta$, $q = x+\frac{1}{2}$, $x \geq A$,

$$|(t+\tfrac{1}{2}-i\nu)^2+M^2| \geq q^2+L_\nu^2,$$
$$|(t+\tfrac{1}{2})^2-\alpha^2/4| \geq q^2-d.$$

The first inequality factors the expression into $q+i(\eta-\nu+M)$ and $q+i(\eta-\nu-M)$; the second is the reverse triangle inequality. Furthermore,

$$(1+\tfrac{1}{2A})^2(2a^2+a^4/A^2) \leq \frac{75}{128}\frac{\nu^4}{16K^2} \leq L_\nu^2. \tag{4}$$

Indeed, after division by $\nu^4/(16K^2)$, the upper bound is $(5/4)^2(8K^2/\nu^2+K^2/A^2) \leq 75/128$. Since $q/x \leq 1+1/(2A)$, (4) implies

$$(x^2+a^2)^2q^2 \leq x^4(q^2+L_\nu^2).$$

Using $x \geq 2d$, we conclude

$$|R_\nu(x+i\eta)| \leq \frac{x^4}{q^2(q^2-d)} \leq \frac{1}{(1+1/x)(1+1/(2x))} \leq \left(\frac{x}{x+1}\right)^{3/2}. \tag{5}$$

The last inequality follows from $(1+u/2)^2 \geq 1+u$.

For any real $R \geq A$, define $H_R = \sup_{R\leq x\leq R+1}|Q_\nu(x+i\eta)|$. Iteration on each unit lattice, starting at $x_0 \in [R, R+1)$, gives

$$|Q_\nu(x+i\eta)| \leq H_R\left(\frac{R+1}{x}\right)^{3/2} \quad (x \geq R), \qquad \int_R^\infty |Q_\nu(x+i\eta)|\,\mathrm{d}x \leq 6RH_R. \tag{6}$$

At fixed $R = A$, compact convergence below bounds H_A uniformly for all sufficiently large ν. This supplies an integrable, mass-independent positive-tail majorant.

3 The downward tail and dominated convergence

Let $s \geq 0$ on $t = -N - is$. Fixed-real-part two-sided Gamma bounds follow from Stirling asymptotics and compactness [2]. Write $B_0 = M - \nu$, $C_0 = M + \nu$. The four large numerator factors, the two large-argument ratios, and the two remaining Gamma factors bound (1) by a constant times

$$\frac{v_0^2}{(v_0+s)^{4N+2}}(1+|B_0-s|)^N(C_0+s)^N(1+s)^{2N}\exp\left[-\frac{7\pi s}{2}-\frac{\pi}{2}(|B_0-s|-B_0)\right]. \tag{7}$$

This remains valid across $s = B_0$, where the real part of the potentially small Gamma argument is $N + 1/2 > 0$. The exponential is $e^{-3\pi s}$ for $s \leq B_0$, and $e^{-4\pi s+\pi B_0} \leq e^{-3\pi s}$ afterwards.

For $v_0 \geq 1$, $M \geq 4\nu$, use

$$\frac{M+s+1}{(v_0+s)^2} \leq \frac{M}{v_0^2}+1 = \frac{2}{K}+1, \qquad \frac{v_0^2}{(v_0+s)^2} \leq 1.$$

Also $1 + |B_0 - s| \leq M + s + 1$ and $C_0 + s \leq 2(M + s + 1)$. Consequently

$$|Q_\nu(-N-is)| \leq C_{K,\alpha,N}(1+s)^{2N}e^{-3\pi s}. \tag{8}$$

No ν-dependent polynomial prefactor remains. The short segment $s \in [-\eta, 0]$ is compact and harmless.

At fixed t on $\mathcal{C}$, Gamma-ratio asymptotics give

$$Q_\nu(t) \longrightarrow U(t) := e^{-2\pi it}(K/2)^{2t}\Gamma(\tfrac{1}{2}-t+\alpha/2)\Gamma(\tfrac{1}{2}-t-\alpha/2). \tag{9}$$

The segment from $-N + i\eta$ to $A + i\eta$ is compact; (6) and (8) dominate the two tails. Thus

$$I_\nu \longrightarrow \frac{1}{2\pi i}\int_{\mathcal{C}} U(t)\ \mathrm{d}t = J. \tag{10}$$

The last equality is the exact auxiliary identity derived in the preceding note using both pole families and the standard Bessel connection [3, 4]. At integer α, continue their complete sum, not the separate singular coefficients. Equation (10) now concerns the full normalized Barnes integral. It is not merely compact-kernel convergence.

4 A growing cutoff justifies the integrated correction

Take $R = \log \nu$, eventually larger than A, N, η. On the truncated contour with $x \le R+1$, $s \le R$, the large Gamma arguments remain in fixed safe sectors. A uniform Gamma-ratio expansion for shifts $h = O(\log \nu)$ follows by Taylor expansion of $\log \Gamma$, using Stirling derivative bounds on discs of radius $O(\log \nu)$:

$$\log \frac{\Gamma(z+h)}{\Gamma(z)} = h \log z + \frac{h(h-1)}{2z} + O\left(\frac{|h| + |h|^3}{|z|^2}\right). \tag{11}$$

Log ratios are continued from zero shift. The discs remain a fixed angular distance from the negative-real cut. Applying (11) at $-i\nu/2, b, c$, with shifts $t, -t, t$, respectively, gives

$$\log \frac{Q_\nu(t)}{U(t)} = \frac{4i}{\nu}(t^2 - t) + O_{K,\alpha}\left(\frac{(1+|t|)^3}{\nu^2}\right). \tag{12}$$

The leading powers contribute only $O(|t|/\nu^2)$ beyond (9): $\log b + \log c = 2\log M - i\pi + O(\nu^2/M^2 + 1/M)$. The b, c quadratic terms are $O(|t|^2/M)$. Exponentiating (12), uniformly on this growing truncation,

$$\left| Q_\nu(t) - U(t)\left[1 + \frac{4i}{\nu}(t^2 - t)\right]\right| \le \frac{C_{K,\alpha}}{\nu^2}(1+|t|)^4 |U(t)|. \tag{13}$$

All polynomial moments of $|U|$ are integrable on $\mathcal{C}$.

It remains to verify that the discarded exact tails are small at this *moving* cutoff. Equation (8) gives

$$\int_R^\infty |Q_\nu(-N - is)| \, \mathrm{d}s \le C(1+R)^{2N} e^{-3\pi R} = o(\nu^{-2}).$$

On $[R, R+1] + i\eta$, (12) bounds $|Q_\nu| \le C|U|$. Reflection and Stirling for U give

$$H_R \le C \exp[-2R \log R + C_K R + O(\log R)].$$

Together with (6), the horizontal exact tail is $o(\nu^{-m})$ for every fixed $m > 0$. The auxiliary tails with the polynomial correction enjoy the same required estimates. Thus integration of (13) yields

$$I_\nu = J + \frac{4i}{\nu}\frac{1}{2\pi i}\int_{\mathcal{C}} (t^2 - t) U(t) \, \mathrm{d}t + O_{K,\alpha}(\nu^{-2}).$$

For $\mathcal{D} = (K/2)\partial_K$ at *fixed* α, the auxiliary Bessel equation gives

$$(\mathcal{D}^2 - \mathcal{D})J = \frac{K^2 + \alpha^2 - 1}{4} J.$$

This proves (2). The moving cutoff is essential: the exact kernel need not have finite polynomially weighted moments on its full horizontal tail.

5 Normalized ratio and finite numerical checks

At $\alpha = 0$, $J(K,0) = -K[K_0(K) + i\pi I_0(K)] \neq 0$ for $K > 0$. Since $C_{M,\nu}$ is independent of α, the theorem implies

$$\frac{\mathfrak{B}(K,\alpha)}{\mathfrak{B}(K,0)} = \frac{J(K,\alpha)}{J(K,0)} e^{i\alpha^2/\nu} + O_{K,\alpha}(\nu^{-2}). \tag{14}$$

This is an absolute normalized-ratio error. A relative formulation also needs $J(K,\alpha) \neq 0$. The exponential agrees with the proved first-order term; no all-orders resummation is asserted.

We independently integrate (1) on the fixed corner contour. For $I_\nu^{(1)} = J[1 + i(K^2 + \alpha^2 - 1)/\nu]$, the results are

K	α	ν	$\lvert I_\nu / I_\nu^{(1)} - 1\rvert$	$\nu^2 \lvert I_\nu - I_\nu^{(1)}\rvert$
2	0.7	64	$3.97586\,10^{-4}$	19.59115
2	0.7	128	$9.56063\,10^{-5}$	18.82311
2	0.7	256	$2.37028\,10^{-5}$	18.66140
4	8	128	0.17119939	75063.29
4	8	256	0.04863051	75956.09
4	8	512	0.01261136	76178.45

The large difference between the last-column scales illustrates why a fixed-parameter theorem cannot silently become a joint-K estimate.

The 69 diagnostics comprise 12 exact recurrence checks, 6 mass inequality checks, 36 rational tail-step samples, 6 quadrature refinements, 4 precision/combined contour-extent comparisons, 4 empirical decreases of the first-correction error, and one normalization comparison with the previous original-z contour. Six compact-expansion residuals are also recorded as data. Quadrature uses 18/30 Gauss nodes per segment and 50/65 digits; the largest mass in each family is checked at 85 digits and with $(X, T, \eta) = (36, 18, 1)$ changed to $(44, 24, 0.75)$. These are empirical finite-point checks, not interval certification. The tail proof is the analytic argument (3)–(8).

6 What remains

The previous full-integral gap is closed for fixed K, α on the symmetric $u = v = 0$, imaginary-ω slice. The constants in this proof depend on K, α; in particular A grows like α^2 and N like $|\alpha|$. The next task is explicit uniform control when $\alpha = Ky$ and $K \to \infty$, including branch cancellation and the transverse variables. The physical external cycles, normalization, disconnected baseline, finite-G limit and microscopic holographic dictionary remain open. This is an author self-reviewed proof draft.

References

[1] NIST DLMF, §5.5, Gamma recurrence and reflection, https://dlmf.nist.gov/5.5.

[2] NIST DLMF, §5.11, asymptotic expansions and remainders, https://dlmf.nist.gov/5.11.

[3] NIST DLMF, §10.25 and §10.27, https://dlmf.nist.gov/10.27.

[4] NIST DLMF, 10.34.2, continuation below the cut, https://dlmf.nist.gov/10.34.